

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 5: System Software

Subtopic 5.1: Operating Systems

Concept 5.1.3: Utility Software

Total Questions: 9

Mark Schemes begin on Page 6

Question 1 | Source: 9618_w25_qp_12 (Q8.b.ii)

- 8 A restaurant accesses software and stores data using cloud computing.
- (b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.
- (ii) The image files that are displayed on the touchscreen have been corrupted.
- Identify **two** examples of utility software that could be used to recover the images.
- 1
- 2 [2]

Question 2 | Source: 9618_w24_qp_11 (Q4.a.ii)

- 4 A student uses a laptop to write a program that is saved as a text file.
- (a) The laptop has utility software and an Operating System (OS).
- (ii) Explain the need for back-up software.
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- [2]

Question 3 | Source: 9618_w23_qp_12 (Q8.b)

- 8 A computer has an Operating System (OS).
- (b) The Operating System has utility software.
- The purpose of some utility software is to improve security.
- Identify **one** example of utility software that is **not** intended to improve security.
- Explain why this software is needed.
- Utility software
- Explanation
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- [3]

Question 4 | Source: 9618_w23_qp_13 (Q5.b.ii)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.
- (b) The program is distributed by downloading the source code and its library files from a web server.
- (ii) The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

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Question 5 | Source: 9618_s23_qp_11 (Q3.c)

- 3 A computer has an Operating System (OS).
- (c) The Operating System has utility software including defragmentation software.

Explain how defragmentation can improve the performance of the computer.

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Question 6 | Source: 9618_s23_qp_12 (Q5.d.ii)

- 5 A student has purchased a new laptop.
- (d) The laptop has systems software.

(ii) Describe the purpose of utility software in a computer.

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Question 7 | Source: 9618_w22_qp_12 (Q1.a)

1 (a) Draw **one** line from each utility software to its most appropriate purpose.

Utility software	Purpose
virus checker	to reorganise files so they are contiguous
disk formatter	to scan for malicious program code
backup	to decrease the file size
disk repair	to initialise a disk
defragmentation	to create copies of files in case the original is lost
	to check for and fix inconsistencies on a disk

[5]

Question 8 | Source: 9618_w21_qp_12 (Q7.c)

7 A computer has system software.

- (c) Identify **two** utility programs that can be used to improve the performance of a computer **and** state how they improve the performance.

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[4]

Question 9 | Source: 9618_s21_qp_11 (Q2.a)

- 2 Bingwen's computer comes with an Operating System and utility software.

- (a) Draw **one** line from each utility software to its correct description.

Utility software	Description
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q8.b.ii)

8(b)(ii)	1 mark per bullet point, max 2 marks • Disk repair / disk contents analysis • Back-up software	2
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Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q4.a.ii)

4(a)(ii)	1 mark for each bullet point (max 2) • To allow data to be retrieved / restored when lost • To automatically make a duplicate copy of data... • ...so the user does not have to remember to backup data • To make regular duplicate copies of data	2
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Mark Scheme for Question 3 | Source: 9618_w23_ms_12 (Q8.b)

8(b)	1 mark for the name of the utility software 2 marks for the explanation • Defragmentation software • ... because over time saving and deleting of small files fragments the disk • ... the software makes (individual) files contiguous • ... so access time to the files is improved • ... because head movement is reduced • Disk contents analysis/disk repair software • ... to identify and mark bad sectors • ... to restore corrupted files • ... to recover lost data (due to hardware failure) • File compression • ... to reduce the size of files • ... which saves storage and memory space // by example • ... and reduces transmission time // by example • Disk formatter • ... to prepare a disk for use // set up the file system • ... to partition the disc • ... to delete all the data from the disc	3
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Mark Scheme for Question 4 | Source: 9618_w23_ms_13 (Q5.b.ii)

5(b)(ii)	1 mark for each bullet point (max 3). • Disk needs to be prepared for initial use • Disk needs to be checked for errors • A new file system needs to be generated on the disk • The file allocation table needs to be set up	3
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Mark Scheme for Question 5 | Source: 9618_s23_ms_11 (Q3.c)

3(c)	1 mark each to max 3 • Rearranges blocks of individual files (on the HDD) so they are contiguous // moves the free space together • Accessing each file is faster • ...because there is no need to search for the next fragment / block of the file • ...so less head movement is needed	3
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Mark Scheme for Question 6 | Source: 9618_s23_ms_12 (Q5.d.ii)

5(d)(ii)	1 mark each to max 2 • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	2
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Mark Scheme for Question 7 | Source: 9618_w22_ms_12 (Q1.a)

1(a)

1 mark for each correct line.

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Utility software

Purpose

virus checker

to reorganise files so
they are contiguous

disk formatter

to scan for malicious
program code

backup

to decrease the file size

disk repair

to initialise a disk

defragmentation

to create copies of files
in case the original is
lost

to check for and fix
inconsistencies on a
disk

Mark Scheme for Question 8 | Source: 9618_w21_ms_12 (Q7.c)

7(c)	1 mark for identifying program 1 mark for description, max 2 per program e.g. • Defragmentation • Less time is taken to access files because each one is contiguous so there is less head movement • Virus checker • makes more RAM available for programs to run • ... because it removes software that might be taking up memory / replicating • Disk repair / Disk contents analysis • preventing bad sectors being used because it identifies / marks them • reduces access times by optimising storage • Disk/system clean up • releases storage by removing unwanted / temporary files	4
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2(a)

1 mark for each correct line

4

Utility software

Description

Disk formatter

Scans software for errors and repairs the problems

Defragmentation

Moves parts of files so that each file is contiguous in memory

Back-up

Creates a copy of data that is no longer required

Disk repair

Sets up a disk so it is ready to store files

Scans for errors in a disk and corrects them

Creates a copy of data in case the original is lost